DETAILED DESCRIPTION OF THE INVENTION

Security, Encryption, and Privacy Architecture

In various embodiments, the penile anatomical geometry profiling system incorporates a multi-layered security architecture configured to protect user-specific anatomical data throughout acquisition, processing, storage, and transmission. Security mechanisms may operate at the hardware, firmware, and application layers to minimize unauthorized access or interception of sensitive geometric information.

In certain embodiments, encryption may be performed directly on the device prior to external transmission. Collected radial distance measurements, motion tracking data, reconstructed geometry models, and encoded profile representations may be encrypted using symmetric encryption protocols, asymmetric encryption schemes, hardware-based secure elements, or trusted execution environments. Device-unique cryptographic keys may be provisioned during manufacturing and securely stored within tamper-resistant memory regions.

The system may further implement secure boot mechanisms to prevent unauthorized firmware modification. Firmware integrity verification may be conducted during startup using digital signature validation or hash-based verification routines. Communication between the device and external computing systems may occur over encrypted channels utilizing secure transport protocols, key exchange mechanisms, and certificate validation processes.

In some embodiments, geometry profiles may be segmented into encrypted data blocks, with separate encryption layers applied to raw measurement data and derived structural metrics. Access controls may restrict decryption capabilities to authenticated user devices or authorized processing environments. Biometric authentication, device pairing protocols, or multi-factor authentication may be required prior to profile retrieval.

Privacy-preserving architecture may further include anonymization routines configured to remove personally identifying information from stored geometry datasets. Encoded structural signatures may be generated to enable comparative modeling or compatibility analysis without exposing explicit dimensional measurements. In certain implementations, federated data processing models may be employed such that comparative analytics occur locally on user devices without transmitting raw geometry data to centralized servers.

The security and cryptographic implementations described herein are not limited to any specific encryption standard, authentication protocol, secure element configuration, or deployment architecture, provided that the system maintains confidentiality, integrity, and controlled access to user-specific penile anatomical geometry profiles derived from synchronized time-of-flight and motion measurements.